

MIS

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Roadmap

Background

- ☒ Technology
- ☒ Infrastructure
- ☒ IS and its core components
- ☒ Technology estimates in finance, accounting, marketing, and management
- ☒ Role of Data
- ☒ IS versus IT

Hardware, Software, and Networks

- ☒ Hardware components
 - ☒ Processing
 - ☒ Storage
 - ☒ Peripherals, IoT, and wearables
 - ☒ Mobile
- ☒ Software
 - ☒ Systems software versus Application software (Apps)
 - ☒ Licensing: Perpetual, Subscription, and Open Source
 - ☒ Cloud Computing: IaaS, PaaS, SaaS, AIaaS, QaaS
- ☒ Networks
 - ☒ LAN, WAN, VPN, WiFi, 5G (Cellular)
 - ☒ Topologies: Star, Mesh, Bus
 - ☒ Internet: TCP/IP

Version Control Systems

- ☒ GitHub
 - ☐ using Git Bash
- ☒ Google Docs
- ☐ Dropbox

Excel

- ☐ Data: extraction, formatting, and cleaning
- ☐ Formula and Statistic presentation
 - ☐ Statistical formulas:
 - ☐ General: AVERAGE(), SUM(), MEDIAN(), STDEV(), PERCENTILE(), MIN(), MAX(), SMALL(), LARGE()
 - ☐ Frequency: COUNT(), COUNTA(), COUNTBLANK(), COUNTIF()
 - ☐ Group-wise: COUNTIF(), COUNTIFS()
 - ☐ Utility formulas:
 - ☐ General: LAG()
 - ☐ Date: YEAR(), MONTH(), DAY(), WEEKDAY()
 - ☐ Conditionals: IF(), IFS(), AND(), OR(), ISBLANK(), IFNA()
 - ☐ String: LEFT(), RIGHT(), MID(), LEN(), UPPER(), LOWER(), PROPER(), &, TEXTJOIN(), FIND(), SUBSTITUTE()
- ☐ Aggregation and Search: VLOOKUP, HLOOKUP
- ☐ Search: MATCH
- ☐ DCF Exercise (Quiz 1)

Case Studies

- ☐ Knight Capital
- ☐ Waddell Reed E-Mini Futures
- ☐ Amazon and Toys-R-Us
- ☐ Spotify and Apple Music
- ☐ Target Pregnancy Prediction
- ☐ Capital One Cloud Services
- ☐ Equifax Data Breach
- ☐ Hugging Face Breach - OpenAI

Assigned Case Studies

- ☐ CrowdStrike Outage
- ☐ McDonald's AI Drive Through
- ☐ Twitter/X Infrastructure Under Musk
- ☐ The MGM Resorts Ransomware Attack
- ☐ The Air Canada Chatbot
- ☐ Amazon's AI Recruiting Engine

SQL

- ☐ Data: creation and extraction
- ☐ Querying:
 - ☐ WHERE with Conditionals:
 - ☐ General: AND, OR, NOT, >, <, >=, <=, <>
 - ☐ Missing: NULL
 - ☐ List and range: IN (), BETWEEN *number1* AND *number2*
 - ☐ String:
 - ☐ LIKE [_ (underscore), % (percent)]

- ☐ GLOB [? (question mark), * (star)]
 - ☐ SUBSTR(*variable_name*, *starting_position*, *number_of_characters*)
- ☐ SORT:
 - ☐ ORDER BY [ASC, DESC]
- ☐ Optional:
 - ☐ HAVING
- ☐ Multi-level Querying
- ☐ Computation and transformation
 - ☐ Statistical formulas:
 - ☐ General: AVG(), SUM(), MEDIAN(), PERCENTILE(), MIN(), MAX()
 - ☐ Frequency: COUNT()
 - ☐ Group-wise: GROUP BY
 - ☐ Utility:
 - ☐ General: LIMIT
 - ☐ Conditional cases: CASE WHEN-THEN-ELSE END AS *variable_name*
 - ☐ Single CASE statement
 - ☐ Multiple CASE statements
 - ☐ Nested CASE statements
 - ☐ Creating Tables:
 - ☐ Permanent using CREATE TABLE *table_name* AS
 - ☐ Temporary using WITH *table_name* AS ()
 - ☐ Dropping Tables:
 - ☐ DROP TABLE *table_name*
 - ☐ DROP TABLE IF EXISTS *table_name*
 - ☐ MERGE: LEFT JOIN, RIGHT JOIN, INNER JOIN, FULL JOIN